

THE EPC ENGINEER

EPC MASTERCLASS

LEVEL 1 | FOUNDATION

Engineering • Procurement • Construction • Commissioning • Start-up

Purpose Build a clear, practical understanding of how an industrial facility moves from a business requirement to safe operating service.

Learning focus	What you will be able to do
EPC fundamentals	Explain the purpose and relationship of Engineering, Procurement and Construction.
Project lifecycle	Describe the path from business need through handover and operation.
Information flow	Recognize that EPC execution is iterative and interface-driven.
Terminology	Use essential project, engineering, procurement, safety, construction and commissioning terms.
Practical application	Map a pump or package from process requirement through commissioning.

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Oil & Gas | Petrochemical | LNG | Refinery | Energy | Industrial Projects

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1. What EPC Means

EPC means Engineering, Procurement and Construction. It is a project-delivery approach used to convert an owner's functional requirement into a completed industrial facility.

Simple definition Engineering defines the facility. Procurement obtains the equipment and materials. Construction assembles the facility. Commissioning and start-up make it operational.

1.1 A simple project requirement

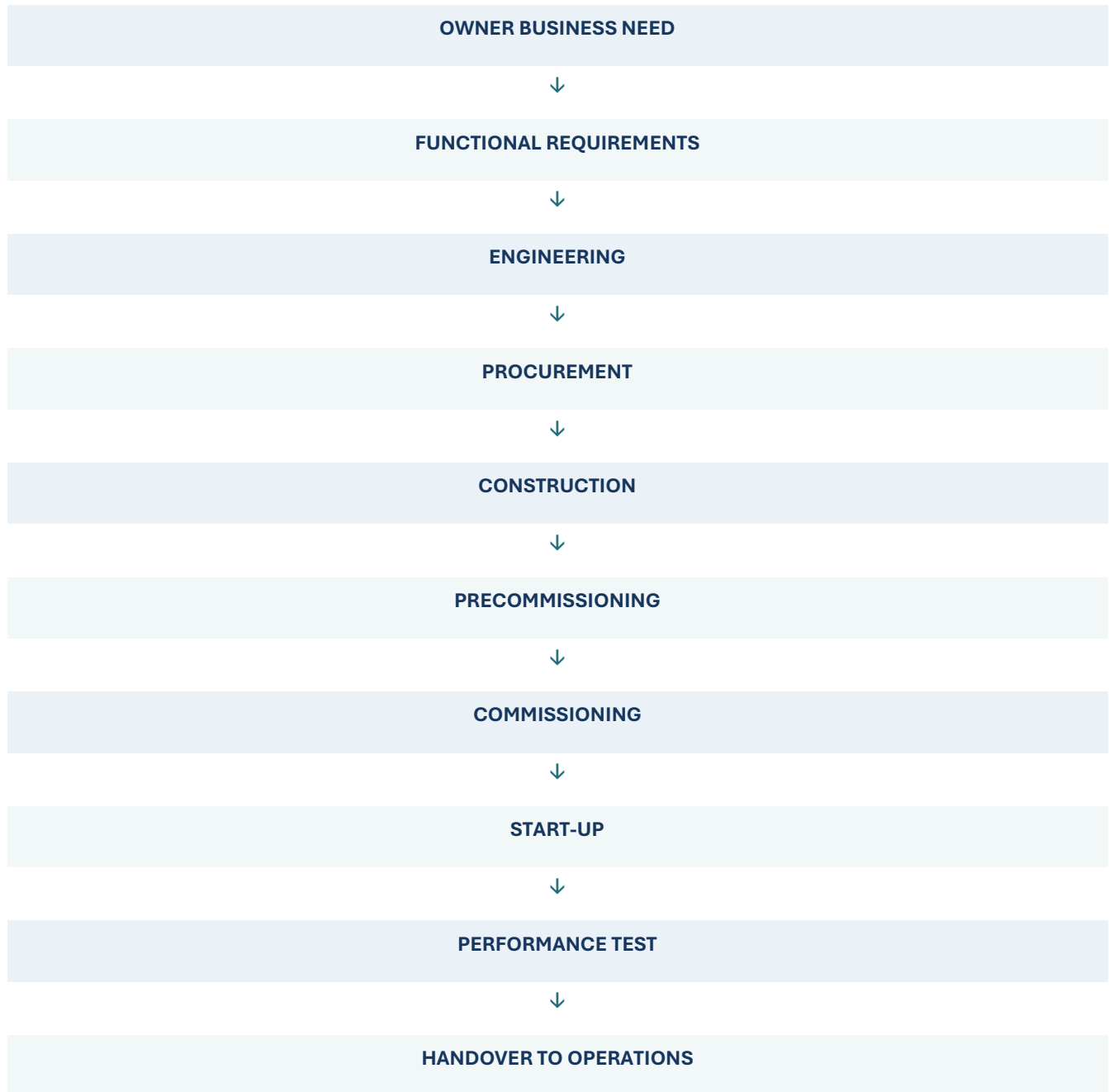
An owner may state: "Build a gas-processing facility that receives raw gas, removes liquids and contaminants, and delivers 100 MMSCFD of sales gas at the required pressure and product specification."

Function	Question answered	Typical result
Engineering	What must be built and how must it perform?	Design calculations, specifications, data sheets, lists and drawings.
Procurement	Who will supply the required equipment and materials?	Qualified vendor, purchase order, approved vendor documents and delivered supply.
Construction	How will the facility be physically assembled?	Installed, inspected and tested plant.
Commissioning	Do systems function safely and correctly?	Energized and functionally tested systems.
Start-up	Can process fluid be introduced and stable operation achieved?	Operating plant progressing toward guaranteed performance.

1.2 House-building analogy

Industrial EPC project	House analogy
Owner requirements	Number of rooms, location, budget and intended use.
Design Basis	Family needs, climate, site conditions and local rules.
Process and system design	How water, power, drainage and spaces will function.
Plot plan	Position of the house and access on the land.
Detail engineering	Architectural, structural, plumbing and electrical drawings.
Procurement	Buying cement, steel, doors, cables and equipment.
Construction	Building and installing the house.
Precommissioning	Checking wiring, water lines and equipment.
Commissioning	Energizing and testing utilities and systems.
Start-up and handover	Occupying the house and receiving keys, manuals and certificates.

2. The EPC Value Chain



2.1 What moves through the chain?

- Requirements become calculations, drawings, specifications and lists.
- Engineering documents become enquiries and purchase orders.
- Purchase orders become vendor drawings, manufactured equipment and materials.
- IFC documents and delivered materials become construction workfronts.
- Completed construction systems become commissioning systems.
- Commissioned systems become an operating facility with records, manuals and warranties.

3. Why Each Phase Is Necessary

3.1 Engineering

Engineering translates performance and functional requirements into a controlled technical definition of the complete facility.

Engineering must determine	Representative outputs
Process function and required capacity	Design Basis, process simulation, HMB, PFD and process description.
Equipment duties and design conditions	Equipment list, process and mechanical data sheets.
Operating, control and safeguarding philosophy	P&IDs, control philosophy, Cause & Effect and shutdown logic.
Materials and physical arrangement	Material selection, plot plans, layouts and 3D model.
Installation requirements	Civil, structural, piping, electrical and instrument drawings.
Purchase requirements	Material Requisitions, specifications, quantities and inspection requirements.

Failure mode Weak or late engineering produces unsuitable purchases, incomplete workfronts, construction queries and rework.

3.2 Procurement

Procurement converts the approved technical requirement into physical equipment, bulk materials and services at the required quality and delivery date.

Procurement activity	Purpose
Vendor qualification	Confirm capability and eligibility.
Request for Quotation	Obtain technical and commercial offers against a controlled requirement.
Purchase Order	Create the contractual basis for supply.
Vendor-document control	Obtain design and interface information needed by Engineering.
Expediting and inspection	Monitor progress and verify quality.
Logistics and delivery	Move, preserve, store and release materials to site.

Failure mode A technically acceptable item can still delay the project if vendor data, manufacturing, inspection or delivery is late.

3.3 Construction

Construction converts engineering deliverables and purchased materials into the physical facility, under controlled quality, safety and sequence requirements.

- Site preparation and temporary facilities.
- Civil foundations, underground networks and structures.
- Equipment erection and alignment.
- Piping fabrication, erection, testing and reinstatement.
- Electrical and instrumentation installation and testing.
- Progressive system completion and handover to commissioning.

Failure mode Construction without approved drawings, materials, access or predecessor completion causes low productivity and rework.

3.4 Precommissioning

Precommissioning demonstrates that installed systems are complete, clean, pressure-tested, calibrated and ready for functional operation.

Typical activity	What it proves
Pressure testing	Pressure boundary integrity.
Flushing, blowing and cleaning	System cleanliness and removal of debris.
Equipment alignment	Mechanical installation readiness.
Cable continuity and insulation testing	Electrical installation integrity.
Instrument calibration	Measurement accuracy.
Loop preparation and reinstatement	Readiness for functional testing.

3.5 Commissioning

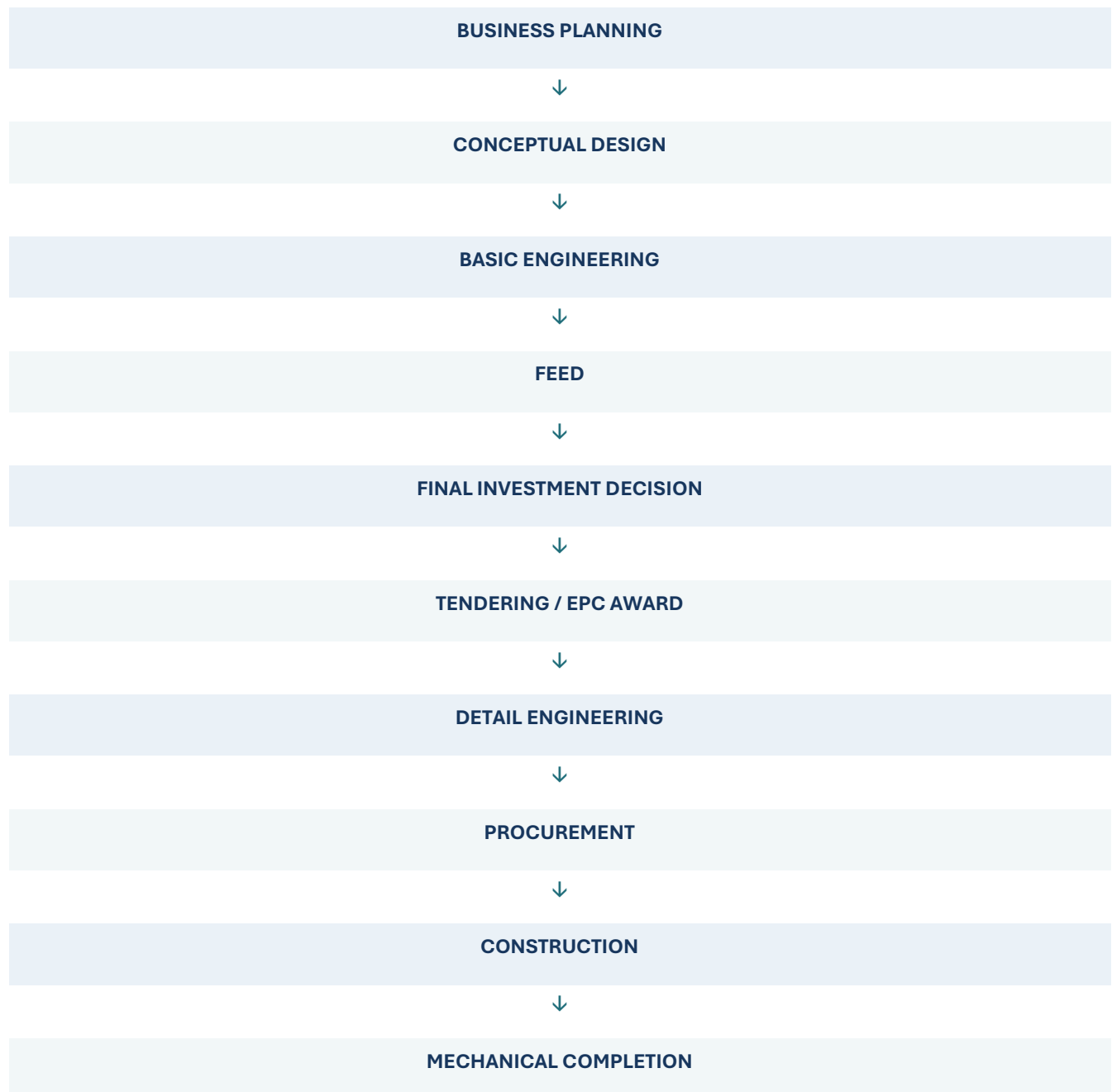
Commissioning demonstrates that each system and the integrated facility function according to approved operating, control and safety philosophies.

- Electrical energization and power distribution checks.
- DCS, PLC, ESD and Fire & Gas functional testing.
- Control-loop, alarm, permissive, trip and sequence testing.
- Utility-system operation.
- Equipment auxiliary-system and no-load operation.

3.6 Start-up, performance test and handover

Stage	Purpose
Start-up	Introduce live process fluids and stabilize operation.
Performance testing	Demonstrate contractual capacity, product quality and other guarantees.
Handover	Transfer accepted systems, records, manuals, spares, warranties and responsibility to the owner/operator.

4. The Complete Project Lifecycle





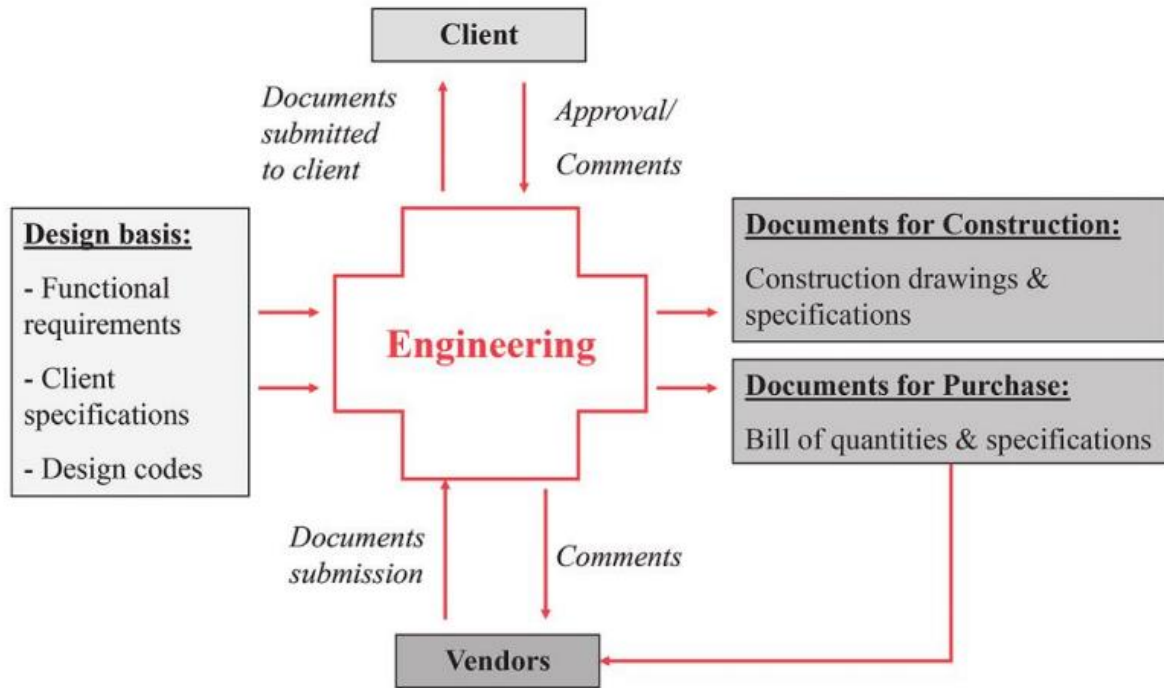
4.1 Stage overview

Stage	Main question	Principal output
Business Planning	Is there a viable business opportunity?	Business case and functional requirement.
Conceptual Design	What broad technical solution is feasible?	Selected concept.
Basic Engineering	What process scheme and major systems are required?	Basic design package.
FEED	Is the scope defined for sanction and EPC tendering?	FEED package and cost/schedule basis.
Tendering	Which bidder provides an acceptable solution?	EPC contract award.
Detail Engineering	Exactly what must be bought and built?	Purchase and IFC documents.
Procurement	Who supplies each item and when?	Equipment, materials and vendor data.
Construction	How will the plant be physically assembled?	Installed facility.
Precommissioning	Is each system complete and clean?	Ready-for-commissioning system.
Commissioning	Does each system function correctly?	Ready-for-start-up system.
Start-up	Can the process operate safely and stably?	Operating facility.
Performance Test	Does the plant satisfy guarantees?	Performance acceptance.
Handover	Can the owner operate and maintain it?	Final handover dossier and accepted facility.

5. Why EPC Is Iterative

The incorrect mental model is a one-way relay: Engineering → Procurement → Construction.

Actual EPC execution involves repeated information exchange among Client, Engineering disciplines, Vendors, Procurement, Construction and Commissioning.



5.1 Examples of iteration

Initial information	Later information	Possible design impact
Estimated equipment dimensions	Certified vendor general arrangement	Plot plan, access, structures and piping routing may change.
Estimated motor power	Vendor confirmed rated power and starting data	Load list, transformer, switchgear, cable and protection may change.
Preliminary equipment weight	Certified operating, test and transport loads	Foundation and lifting studies may change.
Preliminary nozzle orientation	Vendor certified nozzle arrangement	Piping routing and stress design may change.
Preliminary package I/O	Final vendor I/O and Cause & Effect	DCS/ESD configuration, cables and junction boxes may change.
Office design assumption	Site condition or field conflict	RFI, field change and as-built update may be required.

Director's rule Do not only ask whether a document is issued. Ask whether the information is mature enough for the receiving activity.

6. Practical Equipment-Package Example

6.1 Centrifugal pump package

Discipline / function	Key contribution
Process	Fluid properties, flow cases, suction/discharge conditions, differential head, NPSHA and control case.
Mechanical	Pump type, standard, materials, seals, auxiliaries, driver, inspection and reliability requirements.
Piping	Suction/discharge routing, valves, drains, vents, supports, flange rating and nozzle-load management.
Electrical	Motor voltage, starting method, feeder, protection, cable and earthing requirements.
Instrumentation	Pressure, temperature, vibration, alarms, trips, permissives and DCS/ESD interfaces.
Civil	Foundation geometry, anchor bolts, grouting, static and dynamic loading.
Layout	Operating access, maintenance removal, lifting route and spacing.
Safety	Hazardous-area and fire-protection requirements.
Procurement	RFQ, clarification, TBE, PO, vendor documents, inspection, expediting and delivery.
Construction	Foundation, baseplate, grouting, equipment installation, piping, cables and final alignment.
Commissioning	Flushing, lubrication, rotation check, motor solo run, alignment, priming and performance verification.

6.2 Pump package document chain





7. Common EPC Terminology

7.1 Contract and project terms

Term	Meaning
EPC	Engineering, Procurement and Construction.
EPCC	Engineering, Procurement, Construction and Commissioning.
EPCM	Engineering, Procurement and Construction Management.
LSTK	Lump Sum Turnkey.
PMC	Project Management Consultant.
FEED	Front End Engineering Design.
FID	Final Investment Decision.
CAPEX	Capital Expenditure.
OPEX	Operating Expenditure.

Term	Meaning
NTP	Notice to Proceed.
SOW	Scope of Work.
Battery Limit	Defined technical and responsibility boundary.
Greenfield	New facility built on a largely undeveloped site.
Brownfield	Work within or connected to an existing operating facility.
Revamp	Modification or upgrading of an existing plant.
Tie-in	Physical connection to an existing system.
Interface	Boundary where information or responsibility passes between parties.
LLI	Long Lead Item.
Milestone	Significant scheduled project event.
Critical Path	Dependency path controlling overall completion.

7.2 Engineering-document terms

Term	Meaning
BFD	Block Flow Diagram.
HMB	Heat and Material Balance.
PFD	Process Flow Diagram.
P&ID	Piping and Instrumentation Diagram.
Datasheet	Structured record of operating, design and construction requirements.
Specification	Technical requirements governing design, supply or construction.
Calculation Note	Controlled record of design basis, method and result.
Plot Plan	Plant or unit arrangement drawing.
GA	General Arrangement drawing.
MTO	Material Take-Off.
BOM	Bill of Materials.
BOQ	Bill of Quantities.
MR	Material Requisition.
MDR	Master Document Register.
VDR / VDDR	Vendor Document Register / Vendor Data and Document Requirements.
TBE / TBT	Technical Bid Evaluation / Technical Bid Tabulation.

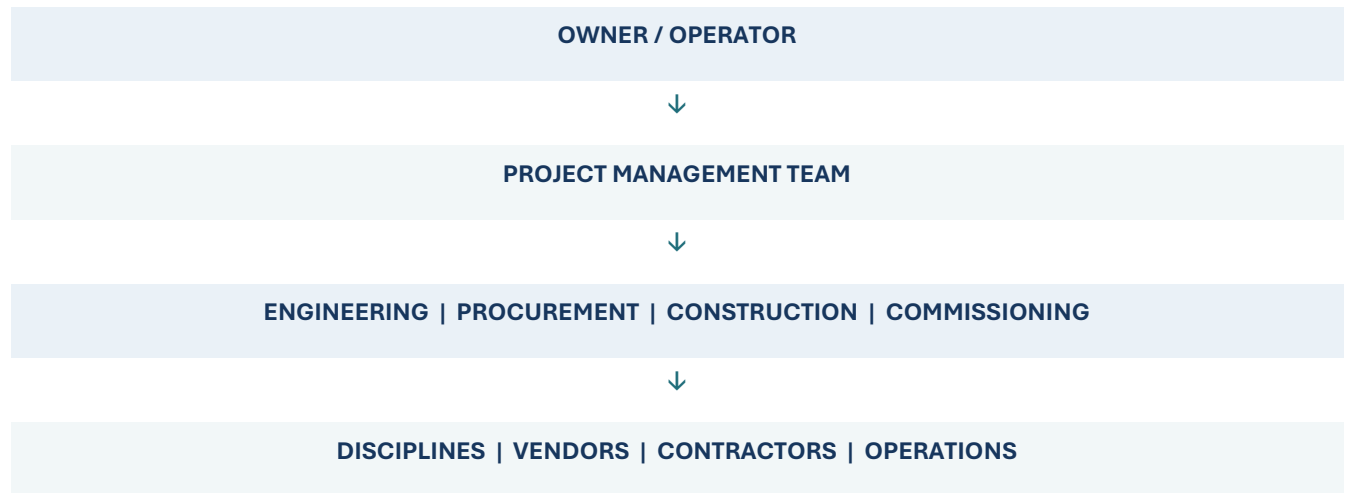
Term	Meaning
TQ	Technical Query.
RFI	Request for Information.
IFR	Issued for Review.
IFD	Issued for Design.
IFC / AFC	Issued or Approved for Construction.
As-built	Final record reflecting the actual installed condition.

7.3 Safety, procurement and completion terms

Term	Meaning
HAZID	Hazard Identification.
HAZOP	Hazard and Operability Study.
LOPA	Layer of Protection Analysis.
SIL	Safety Integrity Level.
QRA	Quantitative Risk Assessment.
DCS	Distributed Control System.
PLC	Programmable Logic Controller.
ESD / PSD	Emergency Shutdown / Process Shutdown.
F&G	Fire and Gas.
C&E	Cause and Effect.
SIF / SIS	Safety Instrumented Function / System.
HAC	Hazardous Area Classification.
RFQ	Request for Quotation.
PO	Purchase Order.
AVL	Approved Vendor List.
ITP	Inspection and Test Plan.
FAT / SAT	Factory Acceptance Test / Site Acceptance Test.
NCR	Non-conformance Report.
ROS	Release for Shipment.
Workfront	Area or system ready for productive construction work.
MC	Mechanical Completion.
RFC	Ready for Commissioning.

Term	Meaning
RFSU	Ready for Start-up.
Punch List	Outstanding defect, incomplete item or record.
Systemization	Division of the facility into commissioning systems and subsystems.
Walkdown	Physical verification of installation and completion.
Reinstatement	Restoration after temporary test arrangements.

8. Project Organization and Ownership



Role	Primary responsibility
Project Director / Project Manager	Overall contract, safety, scope, cost, schedule, risk and client relationship.
Engineering Manager	Multidisciplinary technical integration and engineering delivery.
Procurement Manager	Sourcing, purchasing, expediting, inspection and logistics integration.
Construction Manager	Safe and productive site execution and workfront management.
Commissioning Manager	Systemization, completion, commissioning and start-up readiness.
Project Controls Manager	Planning, progress measurement, cost, forecasting and reporting.
Lead Discipline Engineer	Discipline criteria, deliverables, resources, interfaces and technical quality.
Package Engineer	Single-point coordination of multidisciplinary equipment/package requirements and vendor information.
Document Control	Controlled receipt, issue, revision and distribution of project information.
Operations / Owner representatives	Operability, maintainability, acceptance, training and operational readiness.

9. Three Critical EPC Principles

9.1 Information must arrive before the receiving activity needs it



Schedule consequence Late vendor load and anchor-bolt data can delay foundation design, civil construction and equipment erection.

9.2 “Issued” does not always mean “mature”

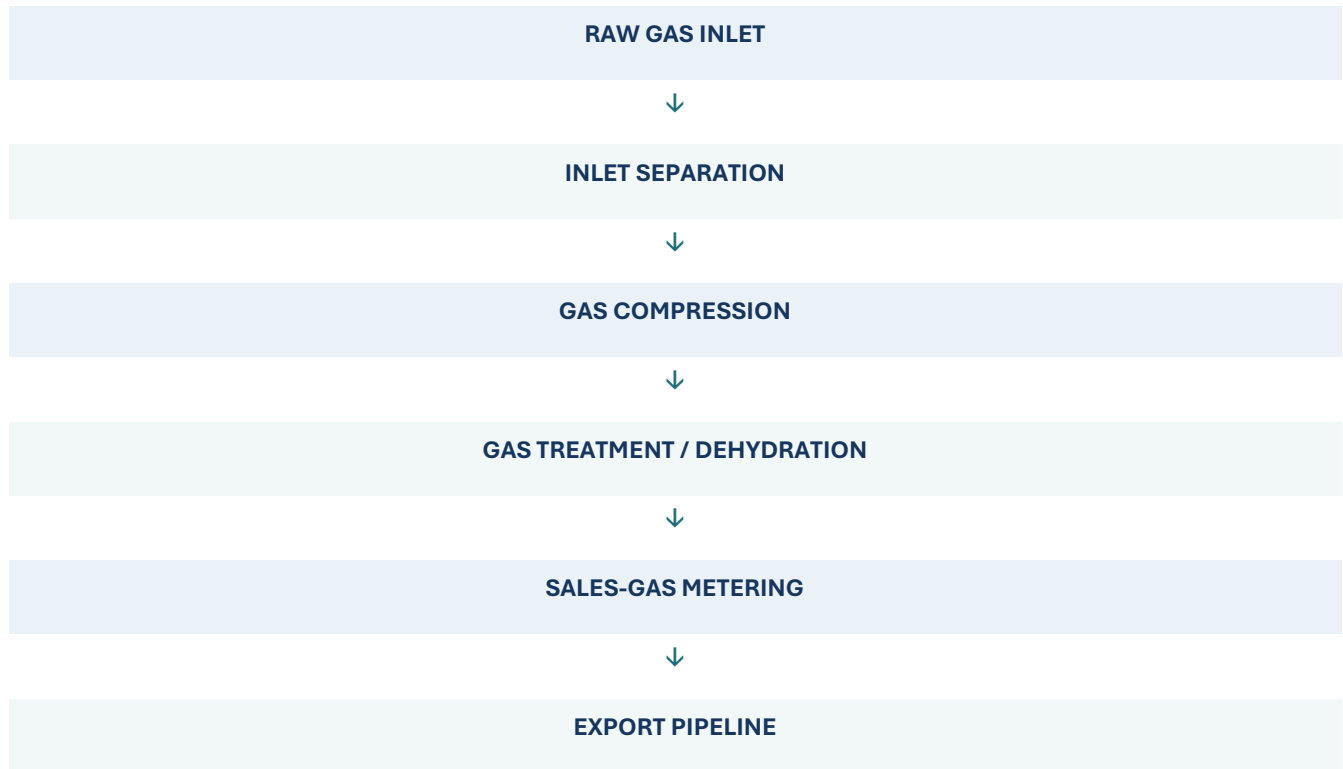
- Check the revision number and issue purpose.
- Confirm client comments, HAZOP actions and vendor data status.
- Verify that the document is approved for the activity being performed.
- Do not use IFR or superseded information for construction unless formally authorized.

9.3 Interface failures cause major project problems

Interface failure	Potential consequence
Wrong motor load transferred to Electrical	Undersized feeder, cable or transformer.
Outdated loads transferred to Civil	Incorrect foundation or anchor-bolt design.
Piping exceeds allowable nozzle loads	Equipment distortion, misalignment or reliability problems.
Trip omitted from instrument scope	Unsafe or non-compliant operating logic.
Maintenance envelope omitted from layout	Equipment cannot be removed or serviced.
Auxiliary excluded from vendor scope	Site change, delay and commercial dispute.
Field change not returned to Engineering	Incorrect as-built record and future operational risk.

10. Foundation Project Example

10.1 100 MMSCFD gas-processing plant



10.2 Parallel liquid and utility systems

Stream / system	Typical destination
Separated condensate	Stabilization, storage or export.
Produced water	Treatment and disposal.
Relief gas	Flare or approved relief system.
Fuel gas	Plant consumers.
Instrument air and nitrogen	Control, purging and inerting.
Electrical power	Generation or grid intake, transformation and distribution.
Drainage	Closed drain, open drain and oily-water handling as applicable.

10.3 Questions before design can begin

- What is the feed-gas composition and variation over field life?
- What are inlet pressure, temperature and flow cases?
- What sales-gas specification is required?
- Must H₂S, CO₂, water, mercury or heavy hydrocarbons be removed?
- What turndown, availability and sparing philosophy apply?
- What utilities and external infrastructure are available?
- What are the battery-limit conditions?
- Is the facility greenfield, brownfield or a revamp?
- What owner specifications, codes and statutory requirements apply?
- What performance guarantees will be contractual?

11. Level 1 Summary and Knowledge Check

Key lesson	Meaning
EPC is an integrated delivery chain	Engineering, Procurement and Construction must be planned as one system.
Commissioning starts in design	System boundaries, access, testing and turnover records must be considered early.
Engineering is iterative	Vendor, routing, site and multidisciplinary information progressively mature the design.
Document status matters	A document must be suitable for its intended use, not merely available.
Interfaces control success	Many serious problems occur between disciplines and organizations.
Information timing drives schedule	The right data must reach the right team before the receiving activity needs it.
Completion is system-based	Construction may be discipline-led, but commissioning and handover progress by systems and subsystems.

Knowledge check

1. Explain EPC in one minute using the house analogy.
2. Describe the difference between precommissioning, commissioning and start-up.
3. Give three examples of vendor data required by other disciplines.
4. Explain why Engineering does not finish before Procurement starts.
5. Describe one interface failure that could delay construction.

12. Interview Questions

6. What is the difference between EPC and EPCM?
7. Why is FEED performed before Detail Engineering?
8. What is the difference between IFR, IFD and IFC?
9. Why does Process Engineering generally start first?
10. How does late vendor data affect the project schedule?
11. What is the difference between Mechanical Completion and Commissioning?
12. What is a battery limit?
13. What makes an item a long-lead item?
14. Why must commissioning planning begin during engineering?
15. What is the purpose of an MDR?
16. What is the difference between MTO, BOM and BOQ?
17. What is the role of a Package Engineer?
18. Why is EPC described as iterative?
19. What is a construction workfront?
20. What documents are required for final handover?

13. Practical Exercise

Prepare a one-page execution map for a chemical-injection package. For each connection, identify the sender, receiver, document, approval, schedule consequence if late, and technical consequence if wrong.




PURCHASE ORDER

VENDOR DOCUMENT REVIEW

FABRICATION + FAT

DELIVERY

FOUNDATION + INSTALLATION

PRECOMMISSIONING

COMMISSIONING

PERFORMANCE VERIFICATION

Connection	Sender	Receiver	Information / document	Required approval	If late	If wrong
Example: PO → vendor documents	Vendor	Package engineer / disciplines	GA, loads, P&ID, load list, I/O list	Project review code	Downstream design delay	Foundation, layout or interface rework
1						
2						
3						
4						
5						

14. Common EPC Engineer Mistakes

- Using preliminary vendor information as certified final information.
- Reviewing a vendor package only from the mechanical discipline perspective.
- Starting construction without checking revision and issue status.
- Managing construction only by discipline and ignoring system turnover needs.
- Beginning commissioning planning too late.
- Leaving package battery limits and responsibility splits ambiguous.
- Tracking document issuance without tracking the decisions and interfaces inside the documents.
- Confusing equipment delivery with construction workfront readiness.
- Ignoring maintenance, lifting and removal requirements during layout development.
- Failing to return field changes into the controlled as-built record.

15. Industry Best Practices

- Establish a controlled Design Basis before discipline design.
- Develop an MDR with planned issue dates, dependencies and required-by dates.
- Identify long-lead equipment during FEED or early Detail Engineering.
- Maintain an interface register for package, discipline and battery-limit interfaces.
- Use staged design reviews and controlled maturity gates.
- Link vendor-document dates to engineering and construction needs.
- Conduct P&ID, HAZOP, 3D-model, constructability, operability and maintainability reviews at the correct maturity.
- Systemize the plant early and align construction turnover with commissioning priorities.
- Use progressive system handover instead of waiting for total plant completion.
- Maintain one authoritative source for approved engineering information.
- Ensure site queries and changes are technically reviewed and incorporated into as-built records.
- Use responsibility matrices for multidisciplinary vendor packages.